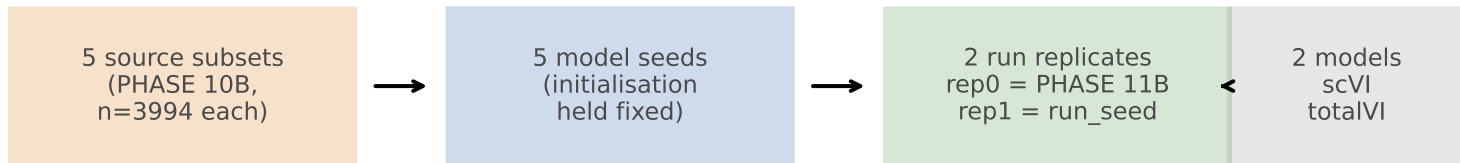


**Figure 1. Replicated  $5 \times 5 \times 2$  experimental design**

**Direction A only: PBMC10k  $\rightarrow$  PBMC5k**



100 observations =  $5 \times 5 \times 2 \times 2$

With  $n=2$  replicates, subset $\times$ seed interaction is separated from pure run-level residual.

**Within each grid cell**

Held fixed:

- exact source subset
  - model\_seed (init)
- architecture / hparams
- query direction / labels

Redrawn via run\_replicate\_seed:

- train/val split, shuffle
- dropout / adapter path

**Identifiability**

PHASE 11B ( $n=1$ ):  
interaction and residual  
confounded ( $\sim 80\%$  lumped)

PHASE 12A ( $n=2$ ):  
 $\hat{\sigma}_{AB}^2$  and  $\hat{\sigma}_E^2$   
separately estimated  
(balanced MoM / EMS)